Source Code and Output

Customer Segmentation using K-Means Clustering (customer_segmentation.py)

Part A: Complete Source Code

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"""
Customer Segmentation using K-Means Clustering
-----------------------------------------------
Dataset : Synthetic "mall customers" data (200 customers) with the same
          schema as the popular Kaggle Mall Customers dataset
          (Age, Annual Income, Spending Score). Replace the data-loading
          block with pd.read_csv("your_file.csv") to use real data.
"""
import os
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.metrics import silhouette_score, davies_bouldin_score
from sklearn.decomposition import PCA

os.makedirs("images", exist_ok=True)
os.makedirs("data", exist_ok=True)

rng = np.random.default_rng(42)

# ---------------------------------------------------------------------
# 1. Build dataset (5 underlying behavioural groups)
# ---------------------------------------------------------------------
def make_group(n, age, income, spend):
    return pd.DataFrame({
        "Age": rng.normal(age[0], age[1], n).clip(18, 70).round(),
        "Annual_Income_k": rng.normal(income[0], income[1], n).clip(15, 140).round(),
        "Spending_Score": rng.normal(spend[0], spend[1], n).clip(1, 99).round(),
    })

df = pd.concat([
    make_group(40, (27, 5),  (30, 8),  (75, 9)),    # young, low income, high spend
    make_group(40, (45, 8),  (28, 8),  (20, 9)),    # mid-age, low income, low spend
    make_group(40, (42, 9),  (85, 10), (50, 8)),    # affluent, moderate spend
    make_group(40, (32, 6),  (88, 10), (82, 8)),    # affluent, high spend (VIPs)
    make_group(40, (55, 7),  (90, 10), (18, 8)),    # affluent, saver
], ignore_index=True)
df.insert(0, "CustomerID", np.arange(1, len(df) + 1))

print("=== DATA OVERVIEW ===")
print(df.shape, "| missing values:", int(df.isna().sum().sum()))
print(df.describe().round(1).to_string())

# ---------------------------------------------------------------------
# 2. Preprocess: scale features (K-Means is distance based)
# ---------------------------------------------------------------------
features = ["Age", "Annual_Income_k", "Spending_Score"]
X = StandardScaler().fit_transform(df[features])

# ---------------------------------------------------------------------
# 3. Choose k: elbow (inertia) + silhouette + Davies-Bouldin
# ---------------------------------------------------------------------
rows = []
for k in range(2, 11):
    km = KMeans(n_clusters=k, n_init=10, random_state=42).fit(X)
    rows.append((k, km.inertia_, silhouette_score(X, km.labels_),
                 davies_bouldin_score(X, km.labels_)))
sel = pd.DataFrame(rows, columns=["k", "Inertia", "Silhouette", "Davies_Bouldin"])
print("\n=== MODEL SELECTION ===")
print(sel.round(3).to_string(index=False))
best_k = int(sel.loc[sel["Silhouette"].idxmax(), "k"])
print(f"\nBest k by silhouette = {best_k}")

# ---------------------------------------------------------------------
# 4. Fit final model
# ---------------------------------------------------------------------
km = KMeans(n_clusters=best_k, n_init=20, random_state=42).fit(X)
df["Cluster"] = km.labels_
# Cross-check with hierarchical clustering
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hc = AgglomerativeClustering(n_clusters=best_k, linkage="ward").fit(X)
print(f"Silhouette  K-Means: {silhouette_score(X, km.labels_):.3f} | "
      f"Hierarchical (Ward): {silhouette_score(X, hc.labels_):.3f}")

# ---------------------------------------------------------------------
# 5. Profile clusters and name them
# ---------------------------------------------------------------------
prof = df.groupby("Cluster").agg(
    Customers=("CustomerID", "count"),
    Avg_Age=("Age", "mean"),
    Avg_Income_k=("Annual_Income_k", "mean"),
    Avg_Spend=("Spending_Score", "mean"),
).round(1)

def name_segment(r):
    hi_inc = r.Avg_Income_k > 60
    hi_sp = r.Avg_Spend > 60
    mid_sp = 35 <= r.Avg_Spend <= 60
    if hi_inc and hi_sp:  return "Affluent Active Spenders"
    if hi_inc and mid_sp: return "Balanced Affluent"
    if hi_inc:            return "Affluent Savers"
    if hi_sp:             return "Young Enthusiasts"
    return "Budget Conscious"

prof["Segment"] = prof.apply(name_segment, axis=1)
prof["Share_%"] = (prof["Customers"] / len(df) * 100).round(1)
print("\n=== CLUSTER PROFILES ===")
print(prof.to_string())
df["Segment"] = df["Cluster"].map(prof["Segment"])

# ---------------------------------------------------------------------
# 6. Visualisations
# ---------------------------------------------------------------------
colors = plt.cm.Set2(np.linspace(0, 1, best_k))
fig, ax = plt.subplots(1, 3, figsize=(16, 4.6))

ax[0].plot(sel["k"], sel["Inertia"], "o-", color="#2b6cb0")
ax[0].axvline(best_k, ls="--", c="grey")
ax[0].set(title="Elbow Method", xlabel="k", ylabel="Inertia")

ax[1].plot(sel["k"], sel["Silhouette"], "s-", color="#2f855a")
ax[1].axvline(best_k, ls="--", c="grey")
ax[1].set(title="Silhouette Score", xlabel="k", ylabel="Score")

for c in range(best_k):
    m = df["Cluster"] == c
    ax[2].scatter(df.loc[m, "Annual_Income_k"], df.loc[m, "Spending_Score"],
                  s=40, color=colors[c], edgecolor="k", lw=0.3,
                  label=prof.loc[c, "Segment"])
ax[2].set(title="Segments: Income vs Spending", xlabel="Annual Income (k$)",
          ylabel="Spending Score")
ax[2].legend(fontsize=7, loc="upper center", bbox_to_anchor=(0.5, -0.18), ncol=2)
plt.tight_layout()
plt.savefig("images/segmentation_plots.png", dpi=160)

# PCA view
pcs = PCA(n_components=2).fit_transform(X)
plt.figure(figsize=(6, 4.5))
for c in range(best_k):
    m = df["Cluster"] == c
    plt.scatter(pcs[m, 0], pcs[m, 1], s=38, color=colors[c], edgecolor="k",
                lw=0.3, label=prof.loc[c, "Segment"])
plt.title("PCA projection of segments")
plt.xlabel("PC1"); plt.ylabel("PC2"); plt.legend(fontsize=7)
plt.tight_layout()
plt.savefig("images/pca_plot.png", dpi=160)

df.to_csv("data/customers_segmented.csv", index=False)
print("\nSaved: images/segmentation_plots.png, images/pca_plot.png, data/customers_segmented.csv")
```

Part B: Console Output (analysis results)

```
=== DATA OVERVIEW ===
(200, 4) | missing values: 0
       CustomerID    Age  Annual_Income_k  Spending_Score
count       200.0  200.0            200.0           200.0
mean        100.5   40.3             63.8            48.5
std          57.9   12.3             29.5            28.3
min           1.0   18.0             17.0             1.0
25%          50.8   31.0             31.8            20.0
50%         100.5   38.5             78.0            52.0
75%         150.2   50.0             88.0            74.2
max         200.0   70.0            110.0            99.0

=== MODEL SELECTION ===
 k  Inertia  Silhouette  Davies_Bouldin
 2  329.747       0.399           1.068
 3  215.313       0.438           0.774
 4  124.619       0.510           0.672
 5   95.894       0.492           0.763
 6   83.319       0.458           0.854
 7   73.894       0.420           0.972
 8   66.608       0.420           0.908
 9   59.921       0.400           0.956
10   55.260       0.404           0.949

Best k by silhouette = 4
Silhouette  K-Means: 0.510 | Hierarchical (Ward): 0.500

=== CLUSTER PROFILES ===
         Customers  Avg_Age  Avg_Income_k  Avg_Spend                  Segment  Share_%
Cluster
0               41     44.5          29.6       18.8          Budget Conscious     20.5
1               69     34.6          85.2       69.9  Affluent Active Spenders     34.5
2               40     27.2          30.1       72.9         Young Enthusiasts     20.0
3               50     55.3          89.3       23.7           Affluent Savers     25.0

Saved: images/segmentation_plots.png, images/pca_plot.png, data/customers_segmented.csv
```

Part C: Visualizations

Figure 1: Elbow method, silhouette scores for k = 2 to 10, and the final segments (income vs spending score)

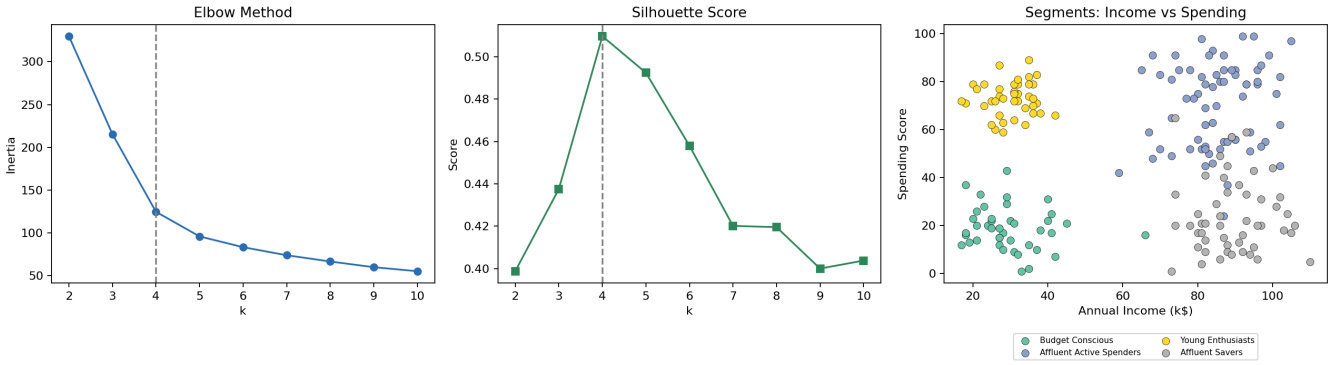


Figure 2: PCA projection of the four customer segments

